Topic 2 – Motion and forces

Students should:		Maths skills
2.1	Explain that a scalar quantity has magnitude (size) but no specific direction	
2.2	Explain that a vector quantity has both magnitude (size) and a specific direction	5b
2.3	Explain the difference between vector and scalar quantities	5b
2.4	Recall vector and scalar quantities, including: a displacement/distance b velocity/speed c acceleration d force e weight/mass f momentum g energy	
2.5	Recall that velocity is speed in a stated direction	5b
2.6	Recall and use the equations: a (average) speed (metre per second, m/s) = distance (metre, m) ÷ time (s) b distance travelled (metre, m) = average speed (metre per second, m/s) × time (s)	1a, 1c, 1d 2a 3a, 3c, 3d
2.7	Analyse distance/time graphs including determination of speed from the gradient	2a 4a, 4b, 4d, 4e
2.8	Recall and use the equation: acceleration (metre per second squared, m/s^2) = change in velocity (metre per second, m/s) ÷ time taken (second, s) $a = \frac{(v - u)}{t}$	1a, 1c, 1d 2a 3a, 3b, 3c, 3d
2.9	Use the equation: (final velocity)^2 ((metre/second)^2, (m/s)^2) – (initial velocity)^2 ((metre/second)^2, (m/s)^2) = 2 × acceleration (metre per second squared, m/s^2) × distance (metre, m) $v^2 - u^2 = 2 \times a \times x$	1a, 1c, 1d 2a 3a, 3c, 3d

Students should:		Maths skills
2.10	Analyse velocity/time graphs to: a compare acceleration from gradients qualitatively b calculate the acceleration from the gradient (for uniform acceleration only) c determine the distance travelled using the area between the graph line and the time axis (for uniform acceleration only)	1a, 1c, 1d 2a 4a, 4b, 4c, 4d, 4e, 4f 5c
2.11	Describe a range of laboratory methods for determining the speeds of objects such as the use of light gates	1a, 1d 2a, 2b, 2c, 2f, 2h 3a, 3c, 3d 4a, 4c
2.12	Recall some typical speeds encountered in everyday experience for wind and sound, and for walking, running, cycling and other transportation systems	
2.13	Recall that the acceleration, g, in free fall is 10 m/s^2 and be able to estimate the magnitudes of everyday accelerations	1d 2h
2.14	Recall Newton's first law and use it in the following situations: a where the resultant force on a body is zero, i.e. the body is moving at a constant velocity or is at rest b where the resultant force is not zero, i.e. the speed and/or direction of the body change(s)	1a, 1d 2a 3a, 3c, 3d
2.15	Recall and use Newton's second law as: force (newton, N) = mass (kilogram, kg) × acceleration (metre per second squared, m/s^2) $F = m \times a$	1a, 1c, 1d 2a 3a, 3b, 3c, 3d
2.16	Define weight, recall and use the equation: weight (newton, N) = mass (kilogram, kg) × gravitational field strength (newton per kilogram, N/kg) $W = m \times g$	1a, 1c, 1d 2a 3a, 3b, 3c, 3d
2.17	Describe how weight is measured	
2.18	Describe the relationship between the weight of a body and the gravitational field strength	1c
2.19	*Core Practical: Investigate the relationship between force, mass and acceleration by varying the masses added to trolleys*	1a, 1c,1d 2a, 2b, 2f 3a, 3b, 3c, 3d 4a, 4b, 4c, 4d

Students should:		Maths skills
2.20	**Explain that an object moving in a circular orbit at constant speed has a changing velocity (qualitative only)**	5b
2.21	**Explain that for motion in a circle there must be a resultant force known as a centripetal force that acts towards the centre of the circle**	5b
2.22	**Explain that inertial mass is a measure of how difficult it is to change the velocity of an object (including from rest) and know that it is defined as the ratio of force over acceleration**	1c,
2.23	Recall and apply Newton's third law **both** to equilibrium situations **and to collision interactions and relate it to the conservation of momentum in collisions**	1a, 1c, 1d 2a 3a, 3b, 3c, 3d
2.24	**Define momentum, recall and use the equation:** **momentum (kilogram metre per second, kg m/s) = mass (kilogram, kg) × velocity (metre per second, m/s)** $\boldsymbol{p = m \times v}$	1a, 1c, 1d 2a 3a, 3b, 3c, 3d
2.25	**Describe examples of momentum in collisions**	1a, 1c, 1d 2a 3a, 3b, 3c, 3d
2.26	**Use Newton's second law as:** **force (newton, N) = change in momentum (kilogram metre per second, kg m/s) ÷ time (second, s)** $\boldsymbol{F = \frac{(mv - mu)}{t}}$	1a, 1c, 1d 2a 3a, 3b, 3c, 3d
2.27	Explain methods of measuring human reaction times and recall typical results	2a, 2b, 2c, 2g
2.28	Recall that the stopping distance of a vehicle is made up of the sum of the thinking distance and the braking distance	1a
2.29	Explain that the stopping distance of a vehicle is affected by a range of factors including: a the mass of the vehicle b the speed of the vehicle c the driver's reaction time d the state of the vehicle's brakes e the state of the road f the amount of friction between the tyre and the road surface	1c, 1d 2b, 2c, 2h 3b, 3c

Students should:		Maths skills
2.30	Describe the factors affecting a driver’s reaction time including drugs and distractions	1d 2b, 2h 3c
2.31	Explain the dangers caused by large decelerations **and estimate the forces involved in typical situations on a public road**	1c, 1d, 2c, 2h, 3b, 3c
2.32P	Estimate how the distance required for a road vehicle to stop in an emergency varies over a range of typical speeds	1a, 1c, 1d 2a 3a, 3b, 3c, 3d
2.33P	Carry out calculations on work done to show the dependence of braking distance for a vehicle on initial velocity squared (work done to bring a vehicle to rest equals its initial kinetic energy)	1c, 1d 2b, 2h 3b, 3c

Use of mathematics

- Make calculations using ratios and proportional reasoning to convert units and to compute rates (1c, 3c).
- Relate changes and differences in motion to appropriate distance-time, and velocity-time graphs, and interpret lines and slopes (4a, 4b, 4c, 4d).
- **Interpret enclosed areas in velocity-time graphs (4a, 4b, 4c, 4d, 4f).**
- Apply formulae relating distance, time and speed, for uniform motion, and for motion with uniform acceleration, and calculate average speed for non-uniform motion (1a, 1c, 3c).
- Estimate how the distances required for road vehicles to stop in an emergency, varies over a range of typical speeds (1c, 1d, 2c, 2h, 3b, 3c).
- Apply formulae relating force, mass and relevant physical constants, including gravitational field strength, to explore how changes in these are inter-related (1c, 3b, 3c).
- Apply formulae relating force, mass, velocity and acceleration to explain how the changes involved are inter-related (3b, 3c, 3d).
- Estimate, for everyday road transport, the speed, accelerations and forces involved in large accelerations (1d, 2b, 2h, 3c).

Suggested practicals

- Investigate the acceleration, g, in free fall and the magnitudes of everyday accelerations.
- Investigate conservation of momentum during collisions.
- Investigate inelastic collisions with the two objects remaining together after the collision and also ‘near’ elastic collisions.
- Investigate the relationship between mass and weight.
- Investigate how crumple zones can be used to reduce the forces in collisions.